

F-CS-WAE: Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoders with Representation-Aware Sampling for Label-Guided Generative Clustering

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Abstract

Generative clustering requires a latent space that is simultaneously discriminative, reconstructive, and samplable. These requirements often conflict: a representation can form clean clusters while decoding poor prior samples, or reconstruct images accurately while providing little class structure. We study this tension through F-CS-WAE, a factorized class-structured Wasserstein auto-encoder that separates a hyperspherical semantic latent $z_c \in \mathbb{S}^{d_c-1}$ from a Euclidean style latent $z_s \in \mathbb{R}^{d_s}$. The semantic component is regularized by class-conditional Spherical Cauchy priors, while the style component is matched to a Gaussian aggregate using MMD. On CIFAR-10, F-CS-WAE achieves $80.87 \pm 0.52\%$ clustering accuracy over three seeds with FID 83.04 ± 0.86 , substantially improving over available VAE, WAE-MMD, VaDE, and single-latent baselines under the repository protocol. On MNIST, F-CS-WAE reaches 99.15% clustering accuracy and strong reconstruction quality, but naive independent sampling $z_s \sim N(0, I)$ yields poor class-conditional generations. We diagnose the cause: the global style aggregate resembles $N(0, I)$, yet style codes remain class-dependent. We then test representation-aware sampling strategies and show that a class-conditional diagonal style prior restores class-consistent generation in a lightweight self-consistency test, improving generated-image self-accuracy from 15% for global Gaussian style sampling to 100% for class-conditional style sampling on a diagnostic subset. The resulting paper claim is sharper than clustering alone: factorized Spherical Cauchy WAEs need sampling rules that respect the learned semantic/style representation. F-CS-WAE is therefore both a strong label-guided clustering model and a diagnostic framework for designing class-conditional generative sampling in factorized latent spaces.

1 Introduction

Auto-encoding generative models promise a compact latent representation from which one can reconstruct inputs, sample new data, and analyze structure. In clustering-oriented settings, however, these goals pull the model in different directions. Reconstruction rewards retaining all information needed to reproduce an image. Clustering rewards removing nuisance variation and preserving class-relevant structure. Prior sampling rewards matching the encoded distribution to a simple distribution from which new latent codes can be drawn. A model that optimizes one of these objectives can easily fail another.

This tension is visible in variational auto-encoders (VAEs) [10], Wasserstein auto-encoders (WAEs) [14], and deep generative clustering methods such as VaDE [8]. VAEs provide a probabilistic latent-variable model, but their Euclidean Gaussian priors do not automatically form class-separated representations. WAEs match the aggregated posterior to a prior, often using MMD, but global matching alone does not impose semantic organization. VaDE introduces a Gaussian mixture prior for clustering, yet it remains Euclidean and does not explicitly separate class semantics from within-class style.

Hyperspherical latent variables offer a natural alternative for semantic organization. If class identity is encoded primarily by direction, then points on a unit sphere can be compared angularly and organized around class centers. Hyperspherical VAEs [4] and Spherical Cauchy VAEs [13] show that distributions on the sphere can be used in deep generative models. The single-latent CS-WAE model in this repository uses this idea: one spherical latent variable is matched to class-conditional Spherical Cauchy priors. It generates relatively clean MNIST samples, but its clustering performance is limited compared with the factorized model studied here.

We introduce F-CS-WAE, a factorized class-structured Spherical Cauchy WAE. The model decomposes the latent representation into two parts. The semantic latent z_c lives on a hypersphere and is aligned with class-conditional Spherical Cauchy priors. The style latent z_s lives in Euclidean space and is regularized toward a Gaussian prior. The architecture is motivated by a simple hypothesis: class identity should be represented by hyperspherical directions, while residual appearance variation should be carried by a separate style code.

The empirical picture is nuanced. On CIFAR-10, the factorized model dramatically improves clustering over available baselines and maintains substantially better sample quality than the older single-latent ResNet variant. On MNIST, it obtains almost perfect clustering and excellent reconstructions, but naive prior samples are worse than those of CS-WAE. A targeted diagnostic shows why: z_s is globally matched to a Gaussian, but remains conditionally dependent on class. Thus, sampling z_c from a class prior and z_s independently from $N(0, I)$ creates latent pairs that the decoder did not learn to map to realistic images.

Rather than treating this only as a limitation, we use it to refine the generative story. If the learned representation is factorized into semantic and style variables, the sampling rule must also be factorized in a representation-aware way. We therefore test class-conditional style sampling rules derived from the encoded style bank. A simple diagonal Gaussian style prior per class gives clean, class-consistent MNIST samples while preserving more diversity than a class mean style code. This paper is therefore written with a stronger but still honest claim: F-CS-WAE improves label-guided clustering, and reliable generation requires conditional style sampling rather than an unconditional style prior.

Contributions.

- We formulate F-CS-WAE, a factorized class-structured WAE with a Spherical Cauchy semantic latent and a Gaussian style latent.
- We introduce a four-phase training schedule that separates reconstruction learning, style regularization, semantic supervision, and class-prior alignment.
- We provide repository-local empirical evidence on CIFAR-10 and MNIST, including multi-seed CIFAR-10 results and a direct comparison against a single-latent CS-WAE variant.
- We diagnose a conditional prior mismatch in the MNIST model, showing that global style MMD is insufficient for reliable class-conditional generation.
- We evaluate representation-aware sampling strategies and show that a class-conditional diagonal style prior repairs class consistency in generated samples in a post-hoc diagnostic.

2 Position in the Field

Auto-encoding generative models. VAEs [10] optimize an evidence lower bound with a KL penalty between approximate posteriors and a latent prior. WAEs [14] instead minimize reconstruction loss plus a discrepancy between the aggregated posterior and a target prior. This makes WAEs attractive for generative representation learning: the regularizer explicitly encourages encoded samples to be drawn from a distribution that is easy to sample. MMD-based WAEs use kernel two-sample distances [6] as the discrepancy.

Generative clustering. Deep Embedded Clustering [15] and VaDE [8] connect latent-variable modeling with clustering. VaDE in particular uses a Gaussian mixture prior, making cluster assignment part of the generative model. F-CS-WAE differs in two respects. First, it is label-guided rather than fully unsupervised: class labels are used during training to structure the semantic prior. Second, it uses hyperspherical semantic geometry rather than a Euclidean mixture.

Hyperspherical and heavy-tailed latent variables. Hyperspherical VAEs [4] motivate distributions on $\mathbb{S}^{d-1}$ when directional geometry is important. Spherical sliced-Wasserstein distances [1] adapt distribution matching ideas to spheres. Spherical Cauchy VAEs [13] further provide an efficient Möbius-transform reparameterization. F-CS-WAE uses this family for the semantic latent, but combines it with WAE-style MMD and label-guided class priors.

Factorization and disentanglement. Factorized latent spaces have a long history in disentanglement. InfoGAN [3] uses mutual information to make selected latent codes interpretable. FactorVAE [9] and β -TCVAE [2] penalize total correlation to encourage independent factors. These works motivate our central diagnostic: if semantic and style variables are intended to factorize, then matching only the marginal style distribution is insufficient. One must also control dependence between style and class or between style and semantic latents.

Where F-CS-WAE fits. F-CS-WAE sits between generative clustering and disentangled representation learning. Its semantic latent is explicitly class-structured, like a supervised version of mixture-prior clustering. Its style latent is intended to capture residual variation. Unlike purely disentangled VAEs, however, F-CS-WAE is evaluated not by interpretability alone but by clustering, reconstruction, and prior sampling. This combined evaluation is important because it exposes a failure mode that a clustering-only table would hide.

3 Method

3.1 Problem Setting

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled image dataset with K classes. The goal is to learn a generative representation that supports:

1. reconstruction: $x \mapsto \hat{x}$ should preserve visual content;
2. clustering: a simple clustering method on semantic codes should recover class structure;
3. sampling: latent samples from the prior should decode into realistic class-conditional images.

We call the setting *label-guided generative clustering*. Labels are used during training to organize the prior, but evaluation of clustering uses K-means and Hungarian matching on latent codes.

3.2 Architecture

F-CS-WAE uses a shared convolutional encoder with two heads:

$$E_\phi(x) = (\tilde{\mu}_c, r_c, \mu_s, \log \sigma_s^2). \quad (1)$$

The semantic direction is normalized,

$$\mu_c = \frac{\tilde{\mu}_c}{\|\tilde{\mu}_c\|_2}, \quad \rho_c = \sigma(r_c)(1 - \epsilon), \quad (2)$$

where $\mu_c \in \mathbb{S}^{d_c-1}$ and $\rho_c \in (0, 1)$. A semantic latent is sampled by a Spherical Cauchy Mobius reparameterization,

$$\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1}), \quad z_c = T_{\mu_c, \rho_c}(\eta_c). \quad (3)$$

The style latent uses a Gaussian reparameterization,

$$\epsilon_s \sim N(0, I), \quad z_s = \mu_s + \sigma_s \odot \epsilon_s. \quad (4)$$

The decoder receives the concatenated latent:

$$\hat{x} = D_\theta([z_c, z_s]). \quad (5)$$

In the current implementation, $d_c = 64$ and $d_s = 128$. The encoder uses a ResNet-18 body, and the decoder is a residual upsampling network. The style log-variance head is initialized near zero and clamped during sampling to avoid numerical overflow.

3.3 Class-Conditional Semantic Prior

For each class k , F-CS-WAE maintains one or more semantic centers

$$m_{k,r} \in \mathbb{S}^{d_c-1}, \quad k = 1, \dots, K, \quad r = 1, \dots, R. \quad (6)$$

The experiments in this draft use $R = 1$. A class prior sample is generated by selecting a center, drawing a uniform spherical noise vector, and applying the same Mobius transform:

$$\eta \sim \text{Unif}(\mathbb{S}^{d_c-1}), \quad z_{c,k}^p = T_{m_k, \rho_p}(\eta). \quad (7)$$

The prior radius is fixed at $\rho_p = 0.7$. Centers are updated by exponential moving average from encoded class means, which stabilizes class anchors while avoiding direct gradient competition with the decoder.

Algorithm 1 F-CS-WAE training loop

Require: Mini-batch (x, y) , encoder E_ϕ , decoder D_θ , class centers m_k

- 1: Encode $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2) \leftarrow E_\phi(x)$
 - 2: Sample $z_c \leftarrow T_{\mu_c, \rho_c}(\eta_c)$ with $\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$
 - 3: Sample $z_s \leftarrow \mu_s + \sigma_s \odot \epsilon_s$ with $\epsilon_s \sim N(0, I)$
 - 4: Decode $\hat{x} \leftarrow D_\theta([z_c, z_s])$
 - 5: Sample class prior semantic latents $z_{c,k}^p$ from Spherical Cauchy centers m_k
 - 6: Compute $\mathcal{L}_{\text{rec}}, \mathcal{L}_{\text{class}}, \mathcal{L}_{\text{agg}}, \mathcal{L}_{\text{style}}, \mathcal{L}_{\text{cls}}$
 - 7: Apply phase weights $\alpha(t), \beta(t), \gamma(t), \eta(t)$ and update ϕ, θ, h
 - 8: Accumulate μ_c by class and update centers m_k with EMA after each epoch
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3.4 Loss Terms

The reconstruction objective is

$$\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x}), \quad (8)$$

with $\lambda_1 = 1$ and $\lambda_{\text{LPIPS}} = 0.1$.

Semantic alignment uses a supervised class MMD:

$$\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_{k=1}^K \text{MMD} \left(\{z_{c,i} : y_i = k\}, \{z_{c,k,j}^p\}_{j=1}^{n_k} \right). \quad (9)$$

An aggregated semantic MMD matches all semantic samples to a mixture of class priors:

$$\mathcal{L}_{\text{agg}} = \text{MMD} \left(\{z_{c,i}\}_{i=1}^b, \{z_{c,r_j,j}^p : r_j \sim \text{Unif}(1, \dots, K)\}_{j=1}^b \right). \quad (10)$$

Style matching uses a multi-scale Euclidean RBF MMD:

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}} \left(\{z_{s,i}\}_{i=1}^b, \{\epsilon_i : \epsilon_i \sim N(0, I)\}_{i=1}^b \right). \quad (11)$$

An auxiliary classifier on μ_c gives

$$\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y). \quad (12)$$

The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} + \eta(t)\mathcal{L}_{\text{cls}}. \quad (13)$$

3.5 Four-Phase Training Schedule

Training proceeds in four phases:

- Phase A, epochs 0–49: reconstruction only.
- Phase B, epochs 50–99: add style MMD and auxiliary classification.
- Phase C, epochs 100–199: add weak semantic class and aggregate MMD.
- Phase D, epochs 200–299: full objective.

The schedule is designed to prevent early regularization from destroying reconstruction. The model first learns a useful decoder, then gradually shapes the semantic and style latents into samplable priors.

3.6 Representation-Aware Sampling

For class-conditional generation, the current model samples

$$z_c \sim p(z_c \mid y = k), \quad z_s \sim N(0, I), \quad \hat{x} = D_\theta([z_c, z_s]). \quad (14)$$

This is the naive factorized generative story. The diagnostic in Section 5 shows that it is not fully achieved on MNIST: z_s remains class-dependent. We therefore also evaluate representation-aware alternatives:

$$z_s = \bar{\mu}_{s,k} \quad (\text{class mean style}), \quad (15)$$

$$z_s \sim N(\bar{\mu}_{s,k}, \tau^2 \text{diag}(\bar{\sigma}_{s,k}^2)) \quad (\text{class-conditional diagonal style}), \quad (16)$$

$$z_s \sim \text{Empirical}\{\mu_s(x_i) : y_i = k\} \quad (\text{style-bank upper bound}). \quad (17)$$

Table 1: CIFAR-10 F-CS-WAE results over three seeds. ACC, NMI, ARI, SSIM are shown as fractions; lower FID and LPIPS are better.

Seed	ACC	NMI	ARI	FID	LPIPS	SSIM	PSNR
0	0.8043	0.6426	0.6259	83.83	0.2126	0.7073	21.88
1	0.8160	0.6607	0.6452	83.45	0.2183	0.6967	21.67
2	0.8059	0.6455	0.6264	81.85	0.2137	0.7060	21.76
Mean	0.8087	0.6496	0.6325	83.04	0.2149	0.7034	21.77
Std	0.0052	0.0079	0.0090	0.86	0.0025	0.0047	0.09

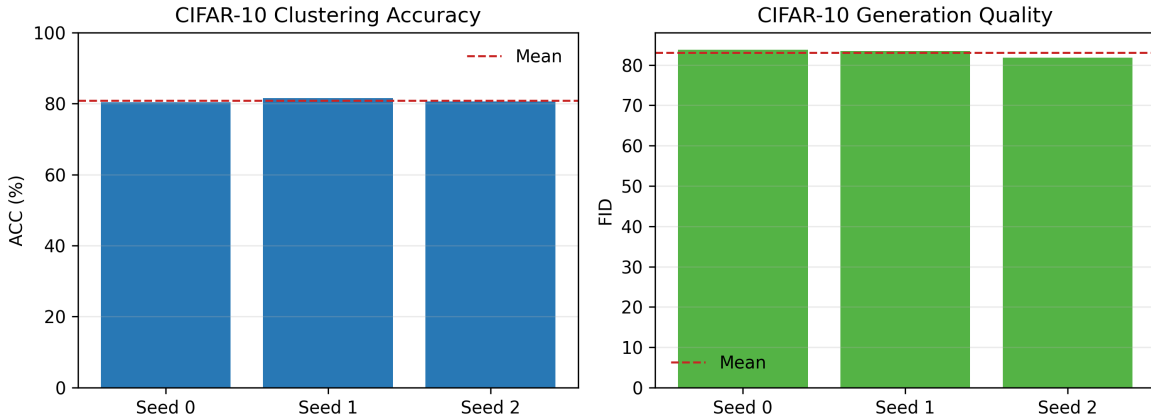


Figure 1: CIFAR-10 clustering and generation metrics across the three completed F-CS-WAE seeds.

The class statistics are estimated from encoded training samples. The scalar τ controls the style temperature. In the MNIST diagnostic, $\tau = 0.25$ gives the cleanest class-consistent samples, while larger temperatures increase diversity but also introduce visible artifacts. This sampling rule does not require retraining the checkpoint, but it changes the generative prior from an unconditional style prior to a conditional factorized prior.

4 Experiments

4.1 Datasets and Evaluation

We report completed repository-local experiments on CIFAR-10 [11] and MNIST [12]. CIFAR-10 is the primary completed multi-seed setting for F-CS-WAE; MNIST is used for detailed analysis because it exposes the style-prior failure mode clearly.

Clustering uses K-means on the semantic means μ_c , followed by Hungarian matching to labels. We report ACC, NMI, and ARI. Generation quality is measured by FID [7]; reconstruction quality uses SSIM, PSNR, and LPIPS [16]. Unless otherwise stated, FID uses 10,000 generated and 10,000 real images.

4.2 CIFAR-10 Main Result

Table 1 reports the completed three-seed CIFAR-10 results. F-CS-WAE is stable across seeds: clustering accuracy varies by only 0.52 percentage points, and FID by less than one point.

4.3 CIFAR-10 Comparison to Available Baselines

Table 2 compares F-CS-WAE to available local baselines. Because not all baselines have three-seed runs, the table explicitly reports the source status. The comparison should be treated as evidence for the current repository protocol rather than as a final benchmark table.

Table 2: Available CIFAR-10 comparison. F-CS-WAE is a 3-seed mean; other rows are completed local seed-0 results unless marked otherwise.

Method	Status	ACC $\uparrow$	NMI $\uparrow$	ARI $\uparrow$	FID $\downarrow$
F-CS-WAE	3-seed mean	0.8087	0.6496	0.6325	83.04
F-CS-WAE full ablation run	completed seed 0	0.8149	0.6622	0.6438	83.59
ResNetAE	seed 0	0.2202	0.1017	0.0493	121.01
VAE	seed 0	0.2220	0.0955	0.0511	179.98
WAE-MMD	seed 0	0.2099	0.0979	0.0469	235.43
VaDE	seed 0	0.1475	0.0427	0.0151	475.87

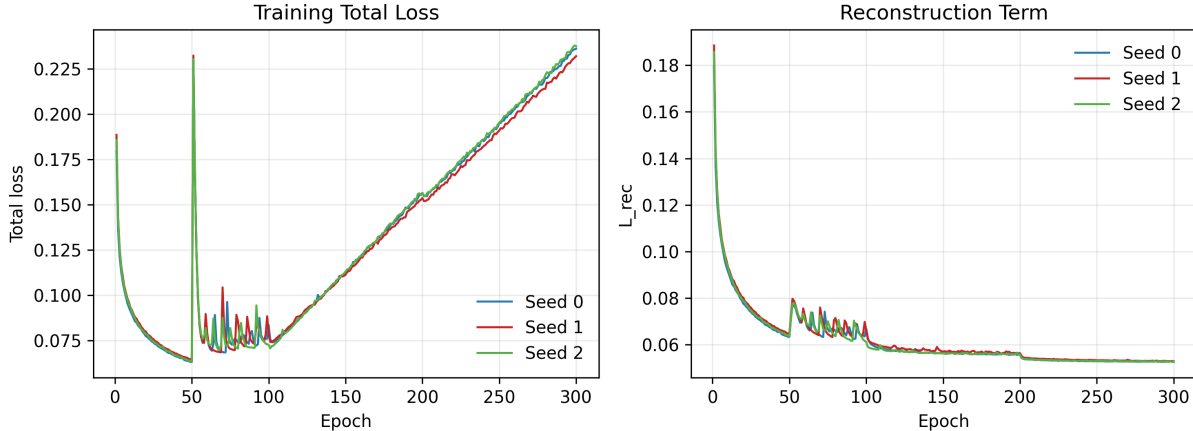


Figure 2: Training curves for the completed CIFAR-10 F-CS-WAE runs. The four-phase schedule keeps training stable after style and semantic regularizers are introduced.

The gap is large: F-CS-WAE improves ACC by roughly 59 percentage points over VAE/WAE-MMD and by roughly 66 points over VaDE in this protocol. The FID gap is also favorable relative to these baselines. This is the strongest current evidence for F-CS-WAE as a paper direction.

4.4 Qualitative CIFAR-10 Structure

Figure 3 shows class-conditional prior samples from the completed CIFAR-10 checkpoint. Figure 4 visualizes the intended factorization: rows with fixed semantics and varying styles, and rows with fixed style and varying semantic centers. These figures are not a substitute for quantitative ablations, but they help inspect whether the decoder uses z_c and z_s in a plausible way.

4.5 MNIST: High Clustering, Poor Naive Prior Sampling

MNIST exposes the central trade-off more sharply. Table 3 compares F-CS-WAE seed 0 to the single-latent CS-WAE seed 0. F-CS-WAE strongly improves clustering and reconstruction metrics, but its FID is worse. This is the key warning for an AAI-27 submission: ACC/NMI/ARI alone would overstate the generative quality of the factorized model.

5 Failure Analysis: Conditional Style Mismatch

The MNIST result raises a question: why does a model with excellent reconstruction and clustering produce poor prior samples? We tested whether the style latent is globally matched to $N(0, I)$ and whether it is independent of class. The answer is mixed.

On 2,048 MNIST test samples encoded by the F-CS-WAE seed-0 checkpoint, the posterior style samples have global standard deviation approximately 0.989, per-dimension variance approximately 0.973, and multi-scale MMD to $N(0, I)$ approximately 0.0017. By a marginal aggregate test, style matching appears successful. However, class-conditional style

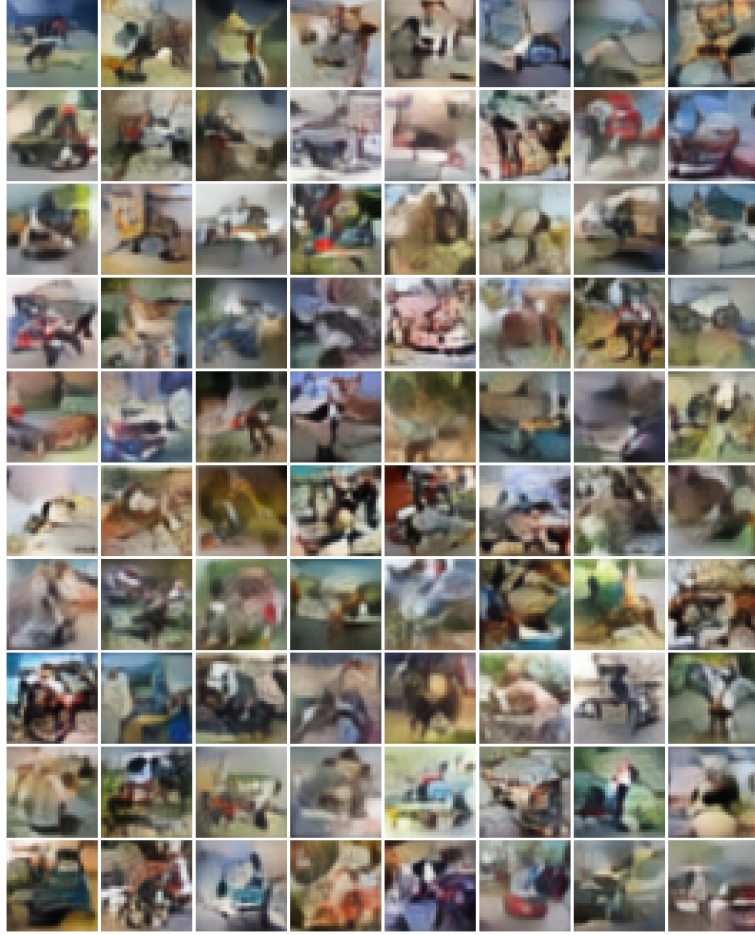


Figure 3: CIFAR-10 class-conditional samples from F-CS-WAE priors. Each row corresponds to a class prior.

means remain far apart: the average pairwise distance between class style means is approximately 6.25, while within-class style norm is approximately 10.25. In other words, z_s still carries class information.

This explains the generation failure. During training, the decoder sees pairs (z_c, z_s) from the same image. If z_s contains class information, then the decoder learns class-specific interactions between z_c and z_s . During generation, the current prior samples z_c from class k but samples z_s independently from the global Gaussian. This creates combinations that may never occur under the encoded posterior.

The conclusion is direct: current F-CS-WAE improves semantic clustering, but its intended factorization is incomplete if generation uses a class-independent style prior. The fix is not to discard the factorized representation, but to sample style in a way that respects the conditional structure that the model actually learned.

6 Representation-Aware Sampling Results

We next test whether generation can be improved by changing the sampling rule rather than retraining the model. The experiment uses the trained MNIST seed-0 checkpoint. We estimate class-conditional style statistics from encoded training examples, then sample z_c from the learned Spherical Cauchy class prior and vary only the choice of z_s . To obtain a lightweight class-consistency metric, generated images are passed back through the F-CS-WAE encoder and auxiliary classifier; the target class is the class used to sample z_c . This metric is not a replacement for FID, but it is useful for detecting the specific failure mode observed above.

Table 3: MNIST seed-0 comparison. CS-WAE is the single-latent Spherical Cauchy WAE variant; F-CS-WAE is the factorized model.

Method	ACC	NMI	ARI	FID	LPIPS	SSIM	PSNR
CS-WAE	0.8735	0.8863	0.8411	26.95	0.0603	0.8127	16.38
F-CS-WAE	0.9915	0.9758	0.9813	73.99	0.0120	0.9833	26.88

Table 4: MNIST representation-aware sampling diagnostics. Self-ACC is computed by generating images from each class prior, re-encoding them with the trained F-CS-WAE model, and checking whether the auxiliary semantic classifier predicts the intended class. Diversity is the mean per-pixel standard deviation over generated images.

Sampling strategy	Self-ACC $\uparrow$	Confidence $\uparrow$	Diversity $\uparrow$
Global Gaussian $z_s \sim N(0, I)$	0.15	0.934	0.151
Class mean $z_s = \bar{\mu}_{s,k}$	1.00	1.000	0.132
Class diagonal Gaussian, $\tau = 0.25$	1.00	1.000	0.135
Empirical same-class style bank	1.00	1.000	0.182

Table 4 confirms the visual diagnosis. The original global Gaussian style sampler has only 15% self-accuracy: generated images often decode as the wrong class or as ambiguous shapes. A class mean style code gives perfect self-accuracy but collapses within-class variation. The empirical style bank is the most diverse, but it is an oracle-like sampler that reuses encoded training styles. The best paper-facing compromise is the class-conditional diagonal Gaussian with temperature $\tau = 0.25$: it is a true parametric sampler, restores class consistency, and preserves more variation than a class mean while producing cleaner samples than higher-temperature variants.

The key scientific point is that the sampling rule must match the learned representation. The model did learn useful semantic and style coordinates, but the original sampler ignored the conditional structure of the style coordinate. Once style is sampled conditionally on the target semantic class, the same decoder produces coherent class-conditional samples. This supports a refined claim for the paper: F-CS-WAE should be evaluated not only by its latent factorization and clustering performance, but also by whether its sampling distribution respects the learned factorization.

7 Toward an AAI-27-Strength Method

The current draft supports the following claim: factorized semantic/style modeling can greatly improve clustering, but naive marginal style matching is insufficient for reliable class-conditional generation. The post-hoc sampling experiment closes part of this gap by using a class-conditional style prior at generation time. For an AAI-27 submission, the method should integrate this prior into the model and evaluate it with full FID, multi-seed runs, and ablations. We propose three concrete extensions.

7.1 Conditional Style Prior

Integrate the post-hoc class-conditional sampler as the default generative prior:

$$z_s \sim p(z_s \mid y = k). \quad (18)$$

The simplest version uses an EMA Gaussian per class:

$$p(z_s \mid y = k) = N(\bar{\mu}_{s,k}, \text{diag}(\bar{\sigma}_{s,k}^2)). \quad (19)$$

This extension matches the empirical finding that style is currently class-dependent and is directly supported by Table 4. It changes the interpretation: style is no longer strictly class-independent. The paper can still make a strong claim if it frames the model as a *conditional factorized prior* rather than a fully disentangled semantic/style model. The next evaluation step is to regenerate FID images using this sampler instead of the global Gaussian sampler.

7.2 Per-Class Style MMD

If the goal is class-invariant style, then the loss should align each conditional style aggregate to the same Gaussian:

$$\mathcal{L}_{\text{style-class}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}(\{z_{s,i} : y_i = k\}, \{\epsilon_j : \epsilon_j \sim N(0, I)\}). \quad (20)$$

This directly targets the observed failure. It is stronger than global style MMD because it prevents class means from separating in style space while still matching the aggregate.

7.3 Style Adversary or HSIC Penalty

A complementary approach is to penalize dependence between z_s and class labels. A gradient-reversal classifier can try to predict y from z_s , while the encoder tries to make this prediction fail. Alternatively, an HSIC-style dependence penalty [5] can reduce dependence between z_s and labels or between z_s and z_c .

7.4 What the Final AAAI Claim Should Be

A paper based only on ACC/NMI/ARI would be vulnerable because the model is explicitly generative. The final AAAI-level claim should be:

Factorized Spherical Cauchy WAE models improve label-guided clustering, and representation-aware class-conditional style sampling converts the learned semantic/style representation into reliable class-conditional generation.

This claim is stronger than a pure clustering result because it contributes both a model and a diagnosis of why factorization can fail.

8 Discussion

Why CIFAR-10 looks stronger than MNIST. On CIFAR-10, F-CS-WAE improves both clustering and FID relative to the available local baselines. This likely reflects the weakness of the baselines under the same ResNet/protocol and the benefit of having a high-capacity factorized representation. On MNIST, the single-latent CS-WAE prior is already well matched to the simple digit manifold, so the factorized model must justify its extra degrees of freedom. It improves clustering, but the style latent becomes an unregulated route for class information.

Why reconstruction is misleading. The MNIST F-CS-WAE checkpoint has excellent SSIM, PSNR, and LPIPS, yet poor prior samples. This confirms a standard lesson in generative modeling: reconstruction quality is not equivalent to sample quality. A decoder can reconstruct encoded latents that lie on the posterior manifold while failing on samples from the intended prior.

Why the failure is useful. The failure is not merely a bad hyperparameter setting. It identifies a missing conditional constraint in the factorized objective. Global matching $q(z_s) \approx N(0, I)$ does not imply $q(z_s | y = k) \approx N(0, I)$ for all k , nor does it imply $z_s \perp y$. The sampling experiment in Section 6 shows that this gap can be addressed at generation time by sampling z_s conditionally on class. This makes the failure useful: it points directly to a better generative prior.

Limitations of the current draft. The completed evidence is not yet sufficient for a final AAAI-27 submission. CIFAR-10 has strong three-seed results, but MNIST and Fashion-MNIST need full multi-seed factorized runs. Ablations are incomplete. The representation-aware sampler is currently evaluated with visual diagnostics and self-consistency, not with full FID. The next version should integrate class-conditional style sampling into the evaluator, regenerate FID images, and compare against the global Gaussian sampler under the same protocol.

9 Conclusion

We presented F-CS-WAE, a factorized class-structured Spherical Cauchy WAE for label-guided generative clustering. The model separates semantic and style latents, aligns semantics with hyperspherical class priors, and regularizes style toward a Gaussian aggregate through MMD. Completed CIFAR-10 experiments show strong and stable clustering performance with favorable generation metrics relative to available baselines. MNIST diagnostics reveal a deeper issue: marginal style matching can hide class dependence in the style latent, causing naive independent prior samples to decode poorly.

The main lesson is that factorization is not obtained by architectural separation alone. A factorized generative model needs a sampling distribution that matches the conditional structure learned by the representation. The post-hoc class-conditional style sampler is a simple and effective step in that direction: it restores class-consistent MNIST generation in the diagnostic setting while preserving the clustering gains of F-CS-WAE. For an AAI-27-strength submission, the next step is to formalize this sampler as the default conditional prior and evaluate it with full FID, matched baselines, and ablations.

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A Repository Artifacts Used

All quantitative claims in this draft are taken from local repository artifacts:

- CIFAR-10 F-CS-WAE metrics: `runs_f/cifar10/seed_{0,1,2}/metrics.json`.
- CIFAR-10 summary tables and figures: `paper_outputs/f_cs_wae_cifar10/`.
- MNIST F-CS-WAE metrics and images: `runs_f/mnist/seed_0/`.
- MNIST CS-WAE metrics and images: `runs/mnist/seed_0/`.
- MNIST sampling diagnostics: `paper-f-cs-wae/experiments/mnist_sampling/`.

B AAI-27 Submission Checklist

Before submission, the following experiments should be completed:

- Run F-CS-WAE on MNIST and Fashion-MNIST for seeds 0, 1, 2.
- Integrate conditional style prior into the official evaluator and compute full FID.
- Implement per-class style MMD as a training-time regularizer.
- Run ablations: no style latent, no semantic MMD, no style MMD, no auxiliary classifier, conditional style prior, per-class style MMD, style adversary, style dimension sweep.
- Recompute all FID directories after changing sampling.
- Add a final benchmark table with matched seeds for all baselines.
- Replace this preamble with the official AAI-27 author kit.

Style-Semantic Diagnostic: fixed z_c varies style; fixed z_s varies class center

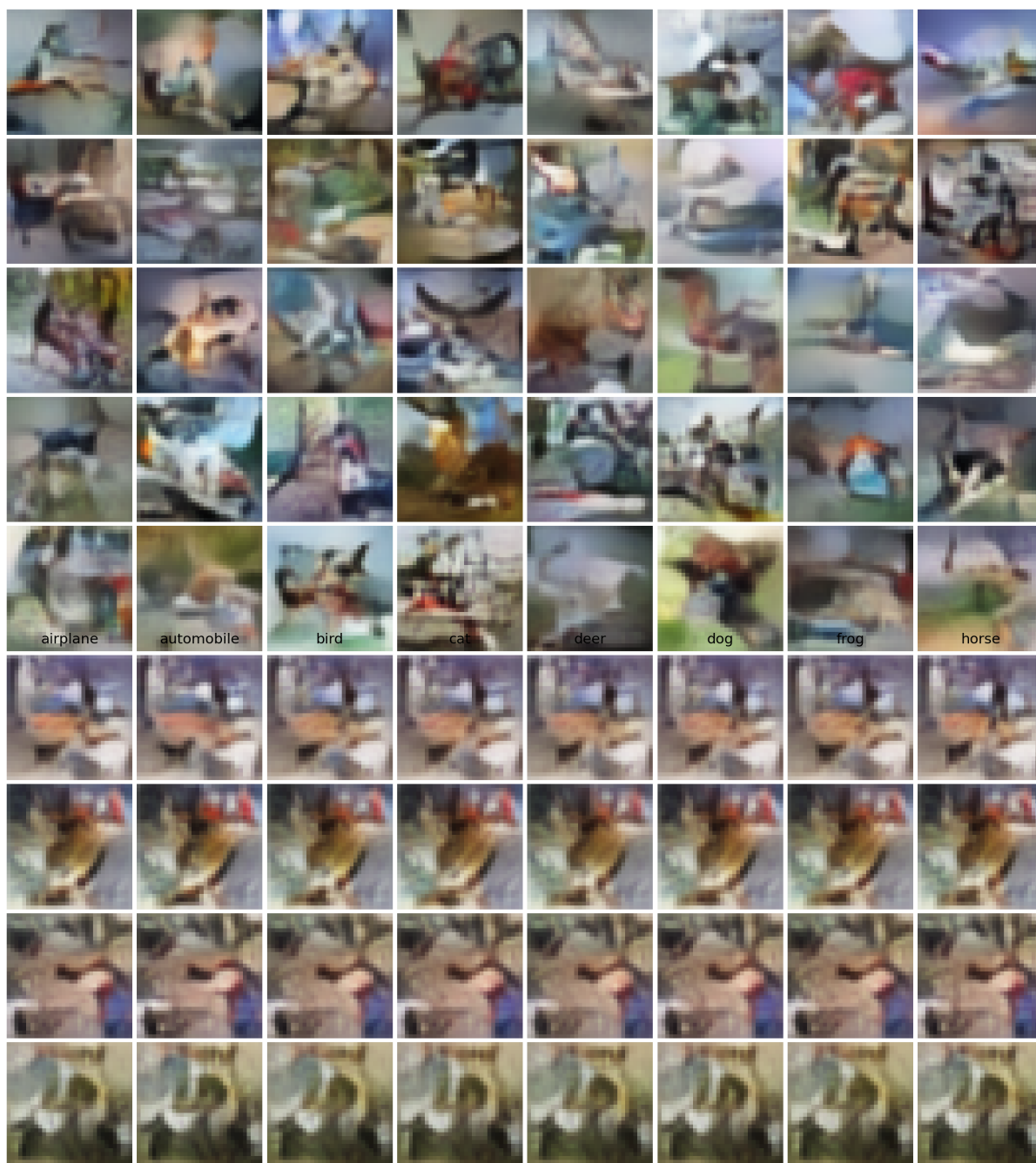
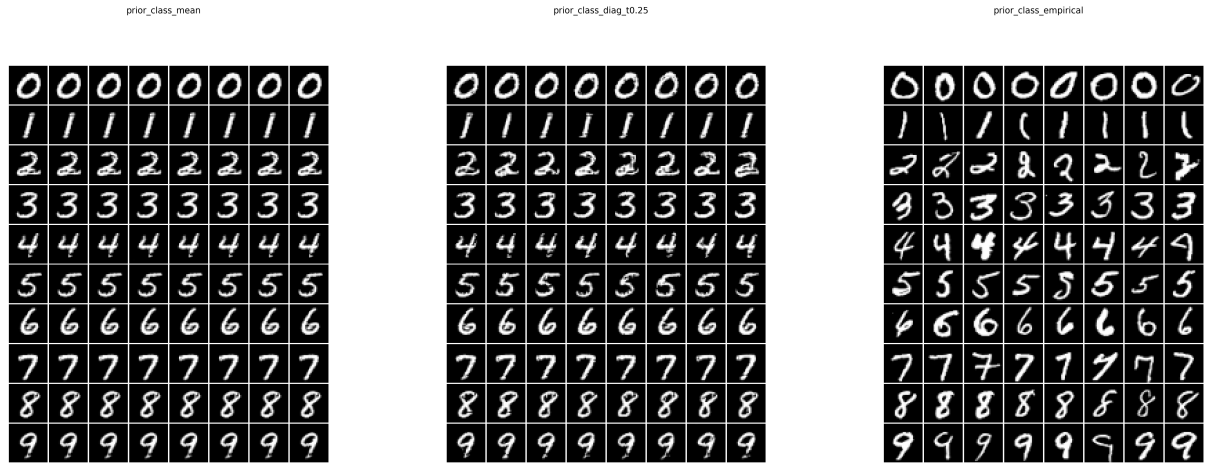


Figure 4: CIFAR-10 style/semantic diagnostic. The figure probes whether varying z_s at fixed z_c , and varying z_c at fixed z_s , produces interpretable changes.



(a) Class mean style.

(b) Class diagonal style, $\tau = 0.25$.

(c) Empirical same-class style bank.

Figure 5: MNIST generation under representation-aware style sampling. The class mean sampler is clean but nearly deterministic; the diagonal class-conditional sampler introduces controlled variation; empirical style-bank sampling is a useful upper bound but less attractive as a standalone generative prior.